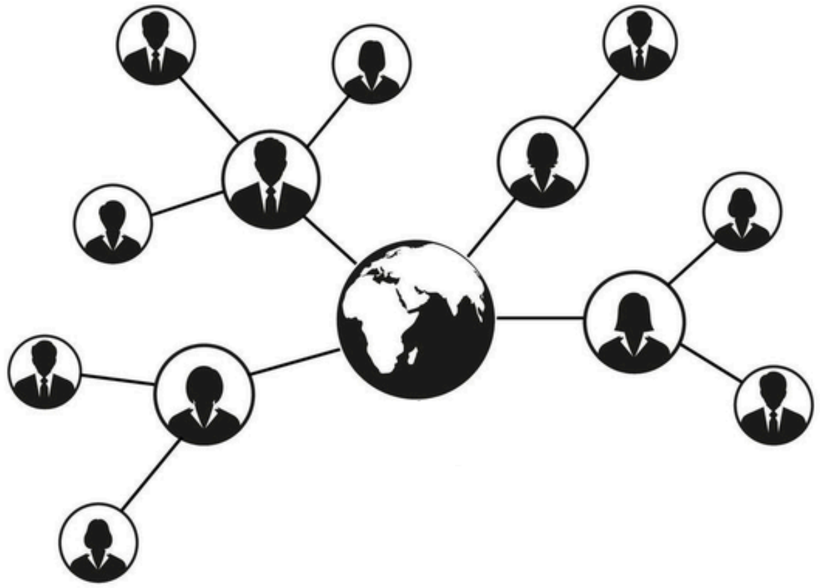
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Distributed Analytics of Global Torrent Activity

A batch + streaming big-data pipeline over public BitTorrent Metadata

Team Members

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The Problem -

- Torrent networks publicly expose how many people share each title.
- Yet piracy is rarely measured systematically across media types or linked to market factors like price, genre and release timing.
- Doing so requires combining millions of torrent records, billions of interest signals and a live stream of swarm statistics from heterogeneous sources — a volume and variety that exceeds single-machine tools.

VOLUME : 2.84M torrents | 1.5B pageview rows
VELOCITY : 25.4M live swarm readings in 10.4 h
VARIETY : 6 heterogeneous sources

(too large and varied for single-machine tools)

APPROACH

A batch + streaming Spark pipeline that unifies these sources into a cross-media piracy index and explanatory models.

Goal: Analyze torrent metadata and answer 6 research queries/questions.

Reference -

- [2005] Pouwelse et al. (IPTPS) - Availability and flash crowds in a single tracker https://pdos.csail.mit.edu/archive/6.824-2007/papers/pouwelse-btmeasure.pdf
- [2005] Guo et al. (IMC) - Torrent popularity decays exponentially https://static.usenix.org/event/imc05/tech/full_papers/guo/guo.pdf
- [2010] Meulpolder et al. (IPTPS) - Public vs. private BitTorrent communities https://www.usenix.org/conference/iptps-10/public-and-private-bittorrent-communities-measurement-study
- [2011] Zhang et al. (IEEE TPDS) - Ecosystem-wide crawl of trackers and torrent sites https://cse.engineering.nyu.edu/~ross/papers/PublicEcosystem.pdf

Data Sources -

- RARBG catalog dump ~ **2.84M torrents** · SQLite dump, Internet Archive
- Live tracker scrapes - **25.4M observations** from 5 public UDP trackers (BEP 15) · 10 Oct 2026
- IMDb datasets - Titles, genres, ratings, cast, episodes · **184.7M rows** · datasets.imdbws.com
- Wikipedia pageviews - Hourly views, **1.15B rows**
- Steam games dataset - **125,854** games: prices, reviews, release dates · Kaggle
- TMDB API - Release dates (digital, theatrical, physical) · **13,462 titles** · themoviedb.org

Research Questions/Queries -

Q1 · Activity

How has upload volume and release quality changed per category?

Q2 · Decay

How fast does a title's swarm shrink as it ages?

Q3 · Leak lag

How soon after cinema and legal home release do films appear?

Q4 · Games

How do price, popularity and game type relate to piracy?

Q5 · Genres

Which film genres are pirated more than their popularity predicts?

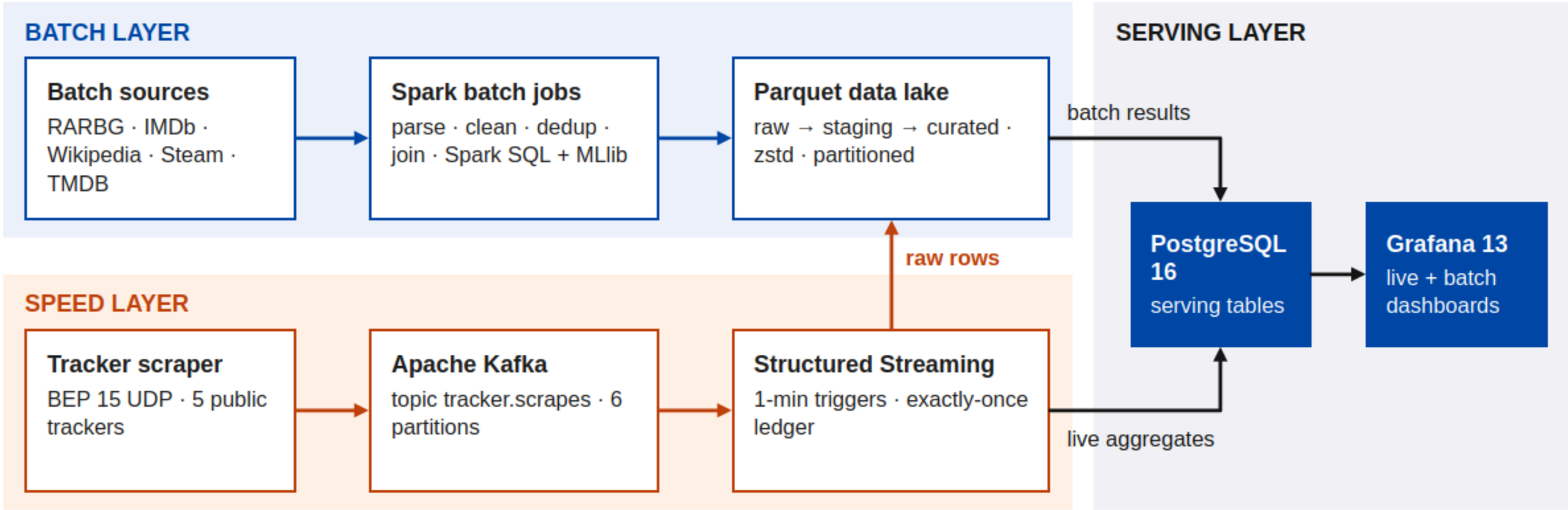
Q6 · Interest

Does public interest (Wikipedia pageviews) track piracy?

Bonus

How swarms change through one day, from the live stream.

Architecture (Lambda) -



- **Kafka, PostgreSQL, Grafana and scrapers in Docker Compose**
- **Spark in local mode on all cores.**

Ingestion and Speed Layer -

INGESTION (OWN CODE)

- **Tracker scraper, BEP 15 UDP:** connect → connection id → scrape up to 74 infohashes per packet
- One loop per tracker: a slow or dead tracker never blocks the others
- **TMDB fetcher:** 20 threads, 35 req/s limit, retries; 10,000 titles in 12 min
- **Bulk dumps:** resumable downloads; RARBG checked against its MD5

SPEED LAYER

- **Two tiers:** 16.5k torrents every ~2 min, 110k every 11–20 min
- Kafka → Structured Streaming, 1-min triggers
- **Two sinks:** raw rows → Parquet; per-minute aggregates → PostgreSQL
- **Exactly-once:** batch id saved in a ledger in the same transaction; replay test passed

Spark Pipeline -

Parsing

- guessit splits 2.84M release names into title, year, season, resolution, codec
- mapInPandas on 12 workers: 60 min
- ANSI mode: try_cast turns bad values into null

Partitioning

- Parquet, zstd, partitioned by date / category / title type
- Layers: raw → staging → curated
- Kafka topic split into 6 partitions

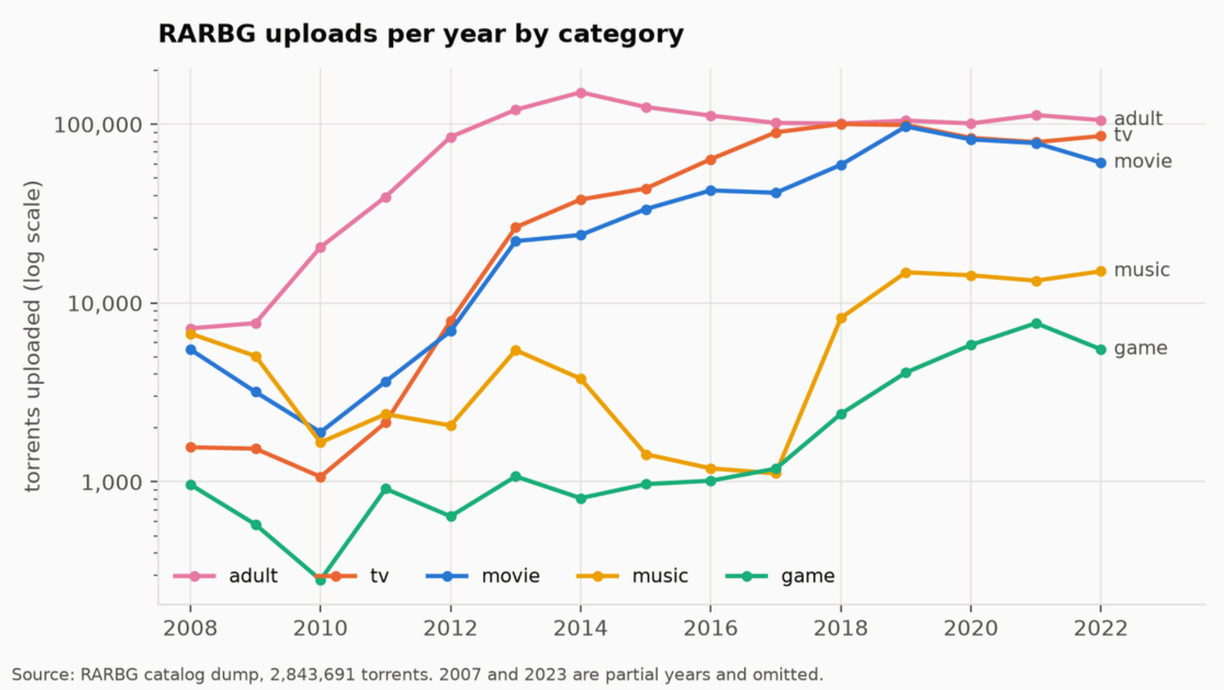
Joins

- Movies/TV → IMDb + TMDB: exact join on IMDb id
- Games → Steam: cleaned names, 79% matched, 34/35 correct
- Titles → Wikipedia: broadcast join over 1.15B rows in 1 min

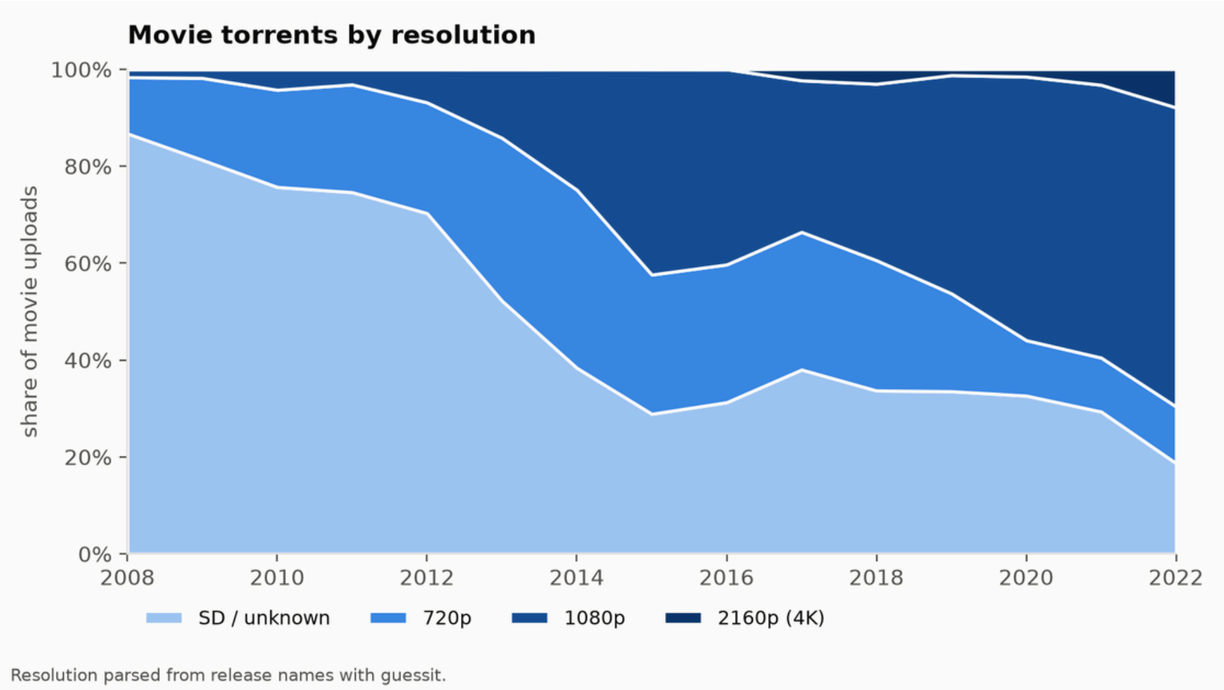
Dedup and cleaning

- Torrents deduplicated by infohash; 25 categories → 6
- Steam: own 40-column schema fixes a broken CSV header
- Wikipedia: 93,017 malformed lines dropped (0.008%)

Q1: Activity Index -



Source: RARBG catalog dump, 2,843,691 torrents. 2007 and 2023 are partial years and omitted.



Resolution parsed from release names with guessit.

~100k / year

TV uploads from 2018 (1,068 in 2010)

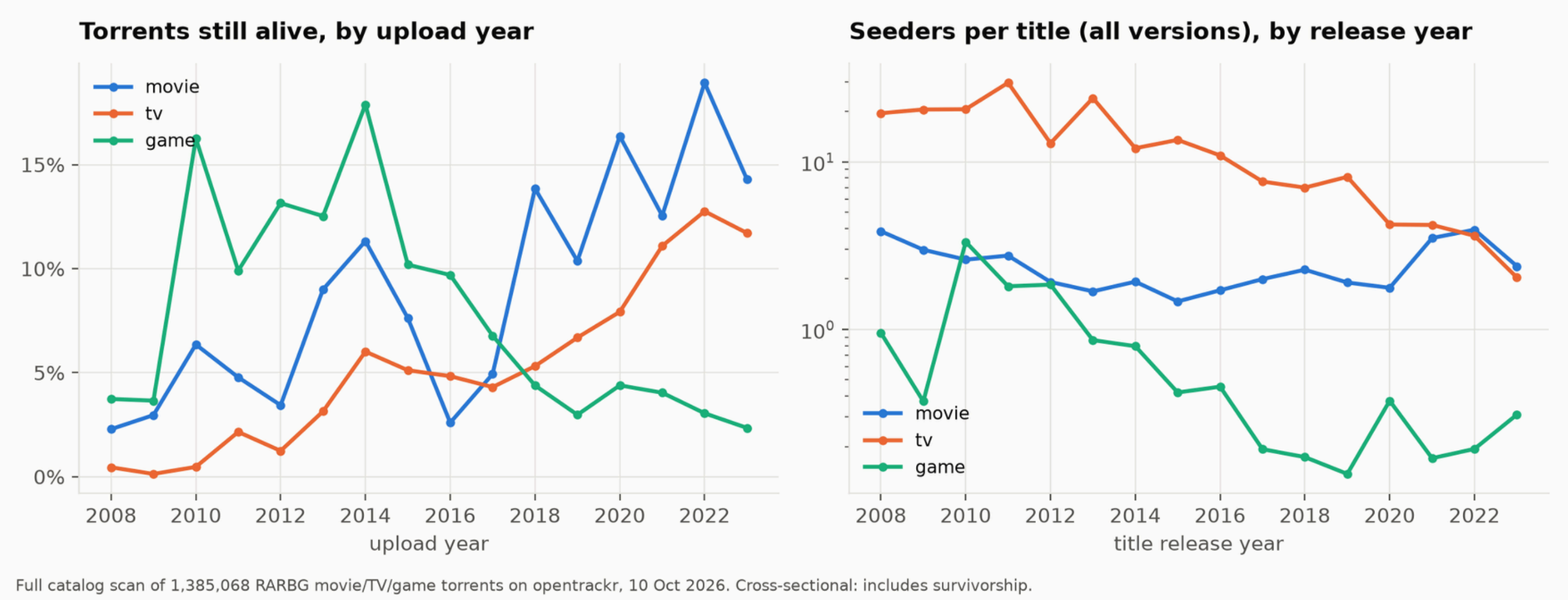
2% → 62%

movie uploads in 1080p, 2008 → 2022

1% → 30%

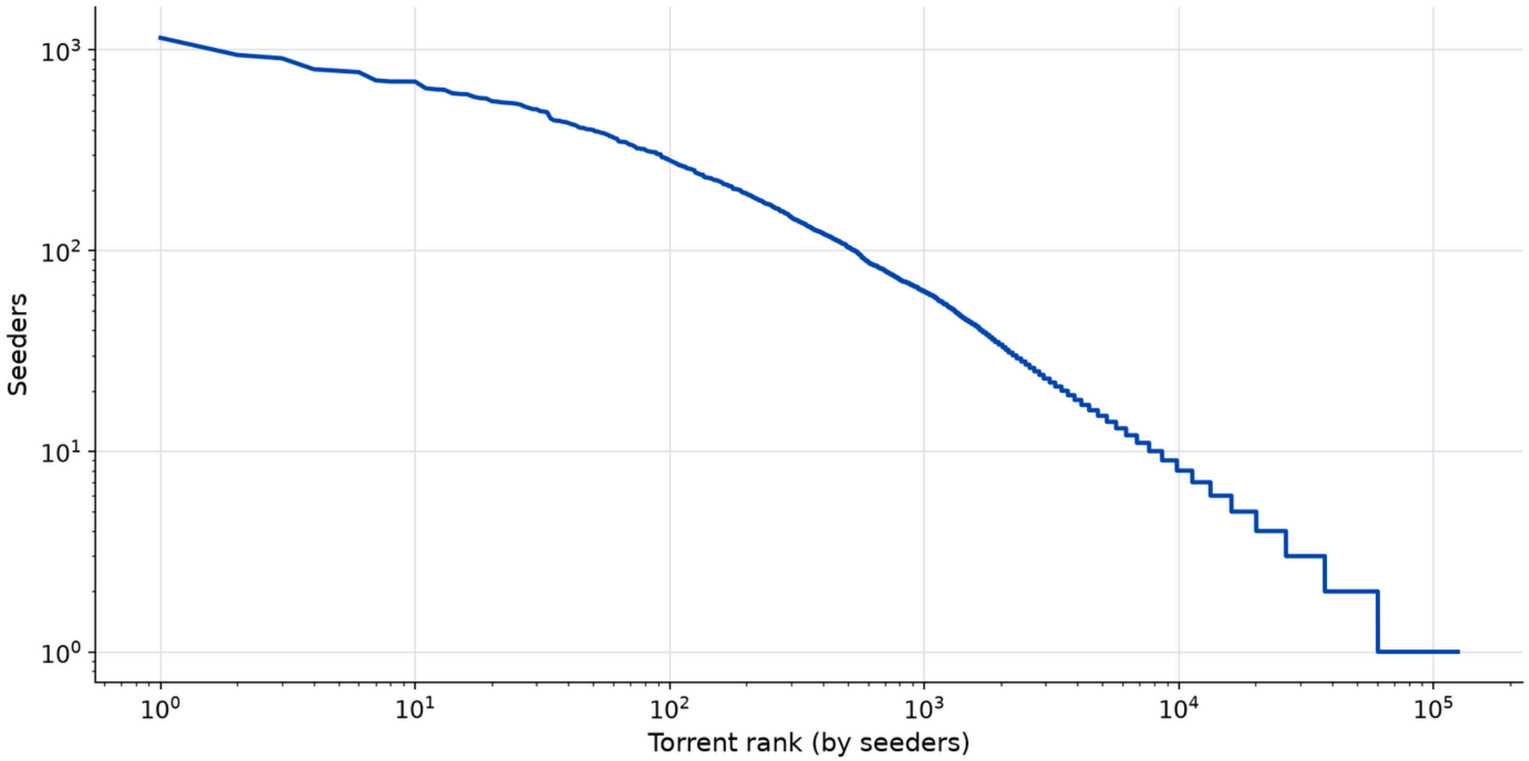
x265/HEVC encoding, 2019 → 2022

Q2: Popularity Decay -



Category	Titles	Best model (AIC)	Meaning
Movie	34,632	power law, age^−0.73	swarm shrinks slowly with age
TV	10,301	power law, +1.65	no decay: series keep adding seasons
Game	10,871	exponential, +16% / year	no decay: classic games persist

Q2: Long Tail and Seeder/Leecher Ratio -



9.0%

of 1.39M catalog torrents still alive

29.9%

of seeders on the top 1% of alive torrents (top 10%: 61%)

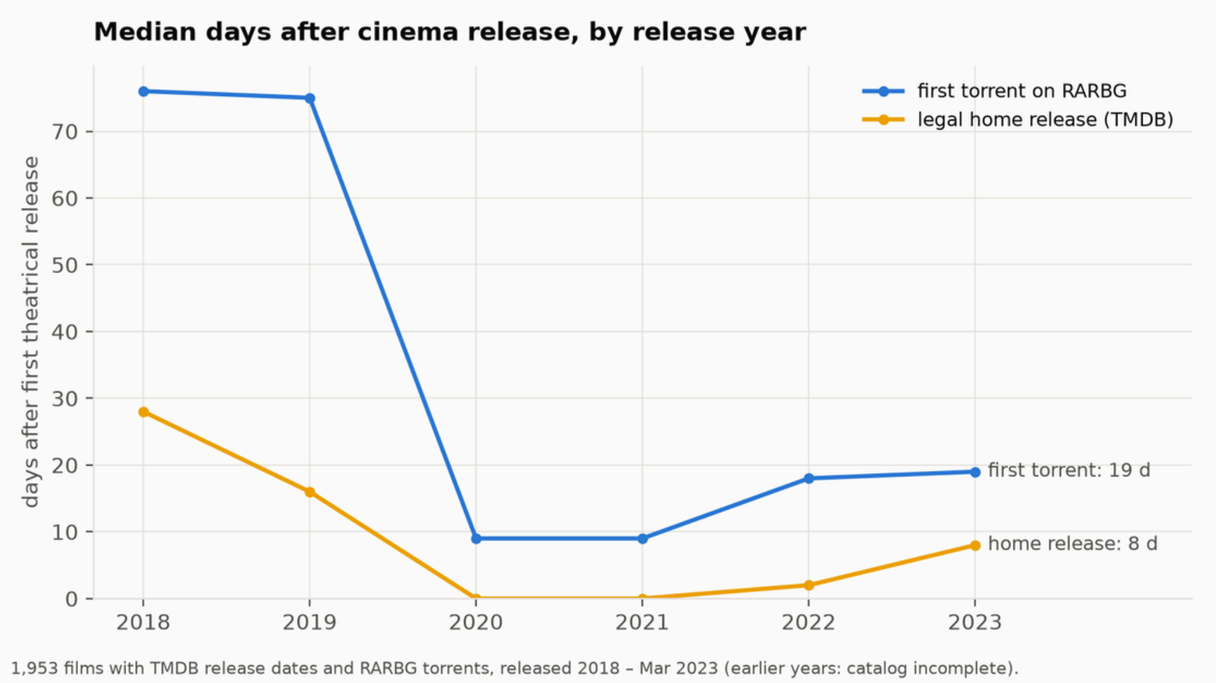
1 seeder

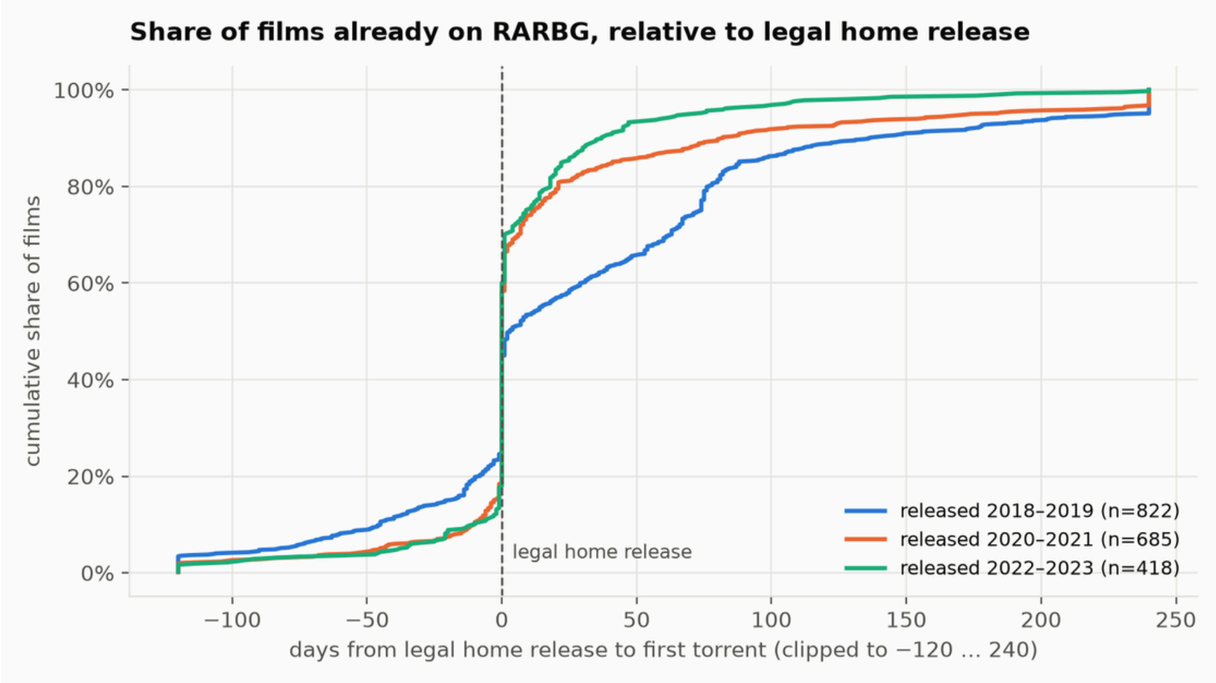
for the median alive torrent

0.42 · 0.35 · 0.24

leechers per seeder: TV · movies · games

Q3: Leak Lag -





76 → 9 days

median cinema-to-torrent lag, 2018 → 2020–21

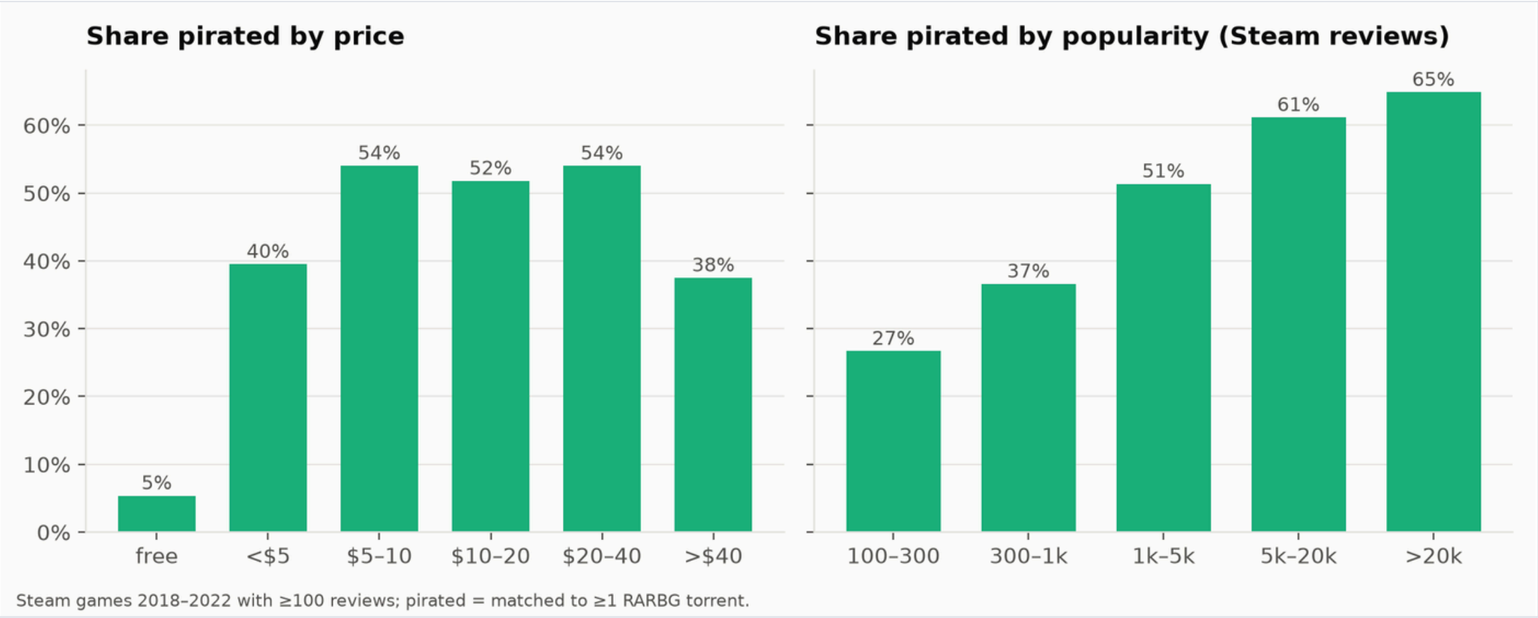
66–70%

of 2020–23 films online by the day after legal home release

1 day

median from TV premiere to first torrent

Q4: Games Regression -



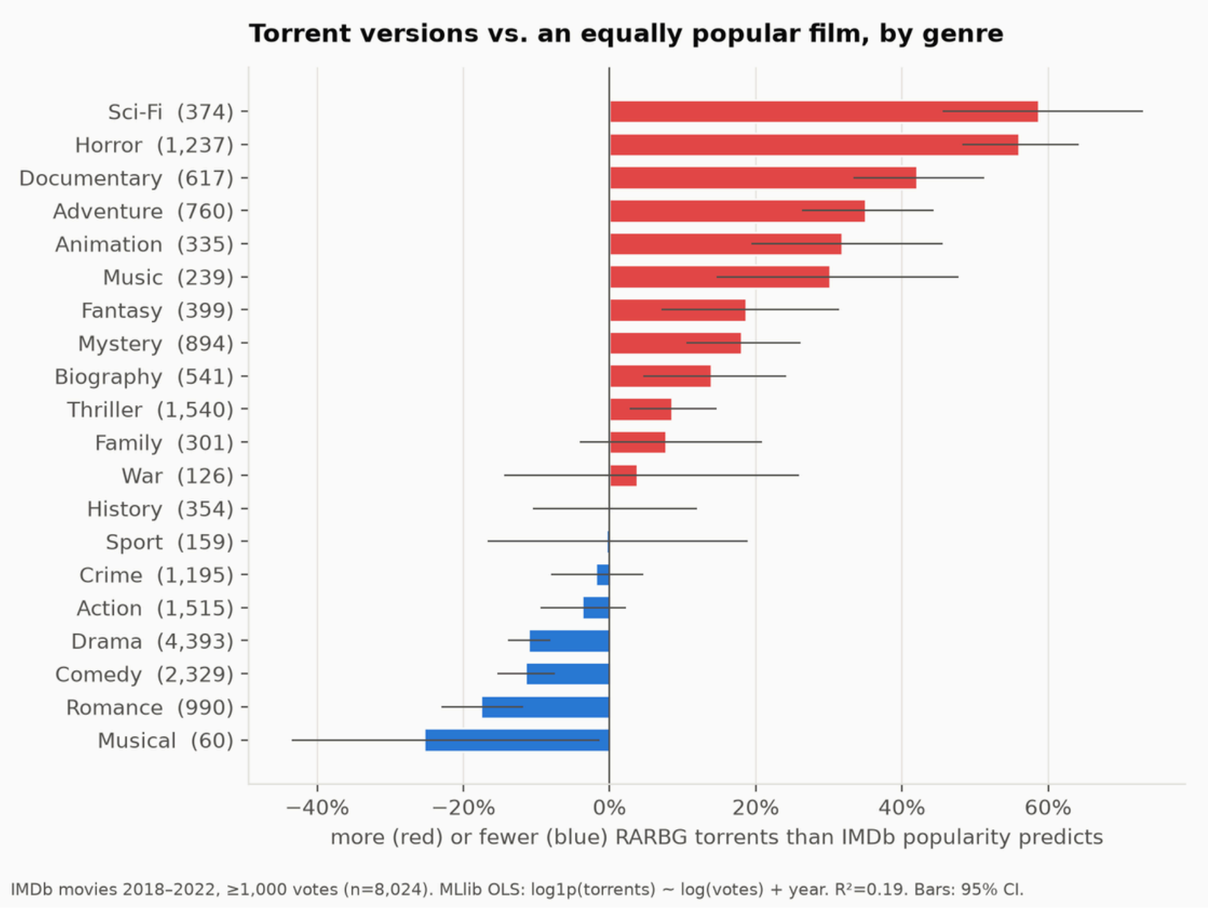
- **Price:** free-to-play 5% · $5–40 52–54% · above $40 38%
- **Popularity:** 27% → 65% pirated as review count rises

LOGISTIC MODEL

AUC 0.83

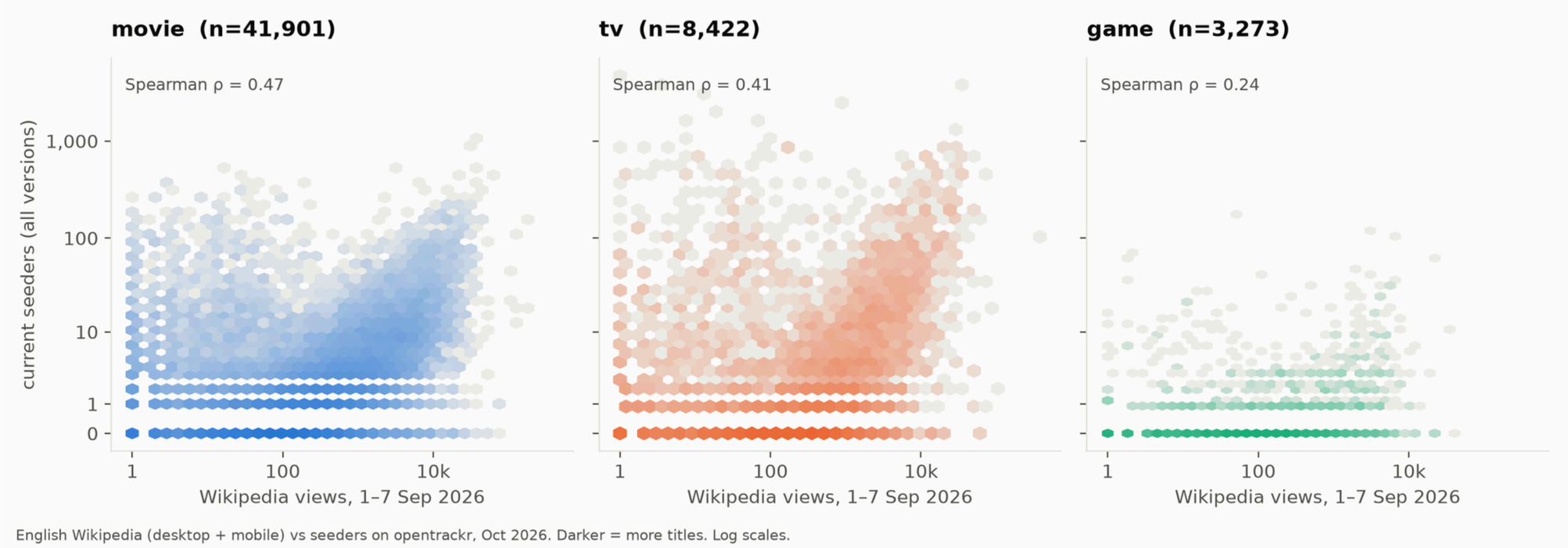
Feature	Odds / SD
Popularity	× 1.84
Single-player	× 1.79
Price	× 1.55
MMO	× 0.77
Casual	× 0.71
Free-to-play	× 0.54
Early Access	× 0.51

Q5: Genres -



Sci-Fi +59%, Horror +56% more versions than equally popular films; Romance −18%, Comedy −12%.

Q6: Interest -



ρ = 0.47

films (41,901)

ρ = 0.41

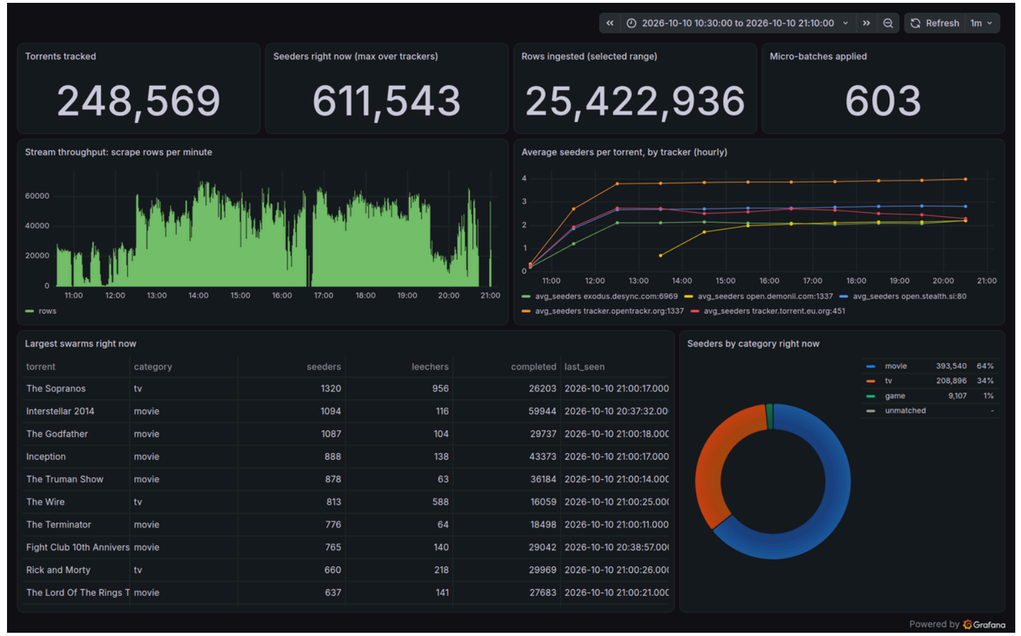
series (8,422)

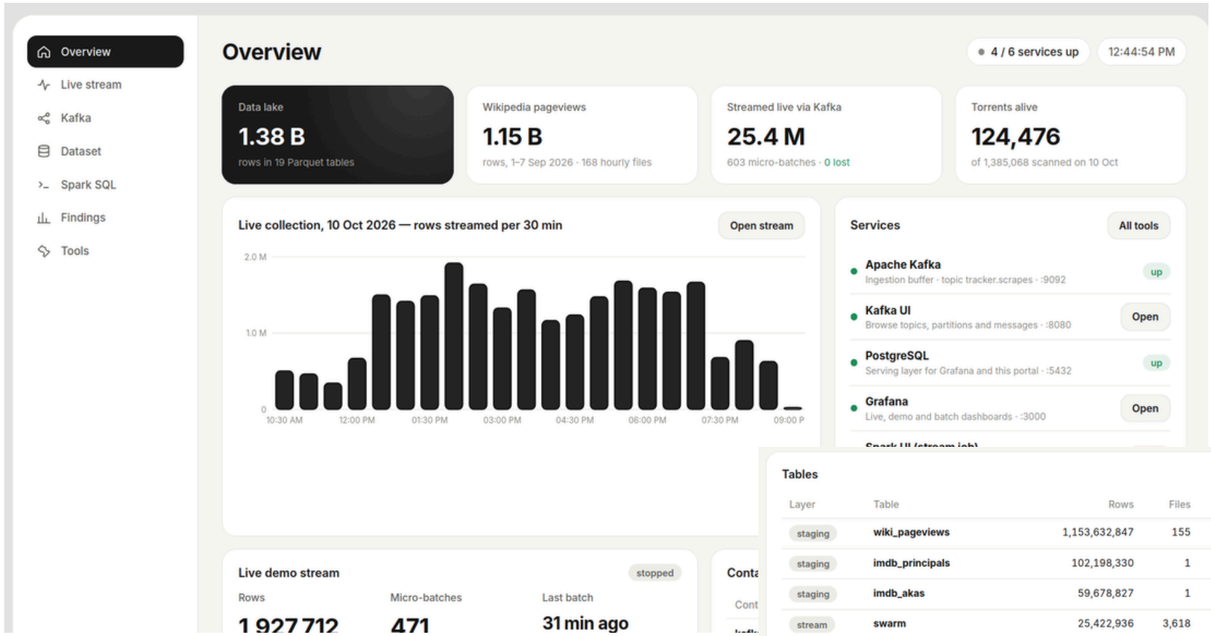
ρ = 0.24

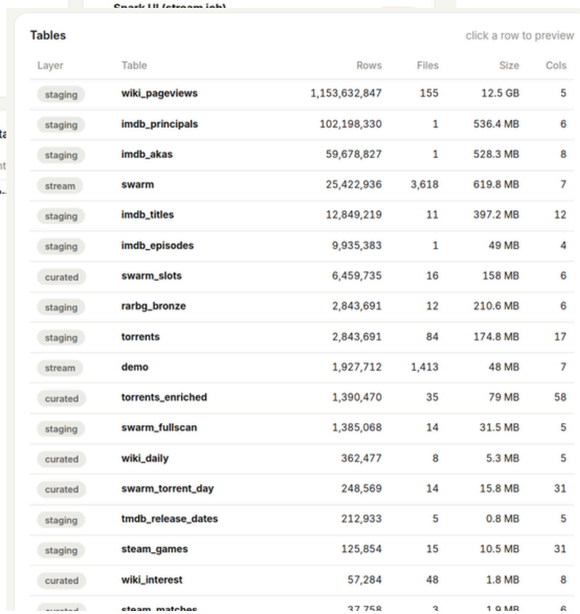
games (3,273)

Spearman ρ, weekly views vs current seeders: moderate, so interest is only a partial proxy for piracy.

Dashboard -







603
micro-batches, exactly-once

3.56M
rows per hour at peak

25.4M
rows streamed in 10.4 h

0 lost
Kafka = ledger = Parquet



Limitations, Ethics and Conclusion -

ETHICS

- Metadata only: no copyrighted content downloaded
- Scraper never requests peer lists: no IP addresses stored
- Only public datasets and public tracker counts

LIMITATIONS

- One site
- Trackers see only part of each swarm
- Decay is cross-sectional; survivorship remains
- Associations, not causal effects

Conclusion: Piracy now follows legal release within a day, film swarms decay as a power law, and price, genre and popularity shape what gets pirated.

Thank You!